

Proximal-to-Distal Energy Transfer in the Golf Swing: Where It Occurs, How It Differs with Skill, and When It Should Happen

A literature synthesis and modeling perspective grounded in the UpstreamDrift
ZTCF/ZVCF counterfactual framework

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Abstract

Proximal-to-distal (P→D) sequencing — the ordered peaking of pelvis, thorax, arm, and club rotational speeds — is the most widely reported organizing principle of the golf downswing. It is usually interpreted as evidence that mechanical energy flows outward along the kinetic chain, and it is often translated into the coaching prescription that energy should be passed distally as early and as completely as possible. This article reviews the mechanics and the empirical literature behind that interpretation and asks a sharper question: *when* during the downswing should the P→D transfer occur? Double-pendulum analyses, delayed-release simulations, and optimal-control results consistently favor accumulating energy proximally first (the arms-plus-club system rotating as a quasi-rigid body), with the transfer to the club deliberately *restrained* early in the downswing and *accelerated* late. Skill-level comparisons show that experts differ from amateurs more in transition timing, segment-speed magnitudes, and transfer efficiency than in the mere presence of a P→D order, and methodological work cautions that peak-ordering alone is a weak proxy for energy flow. We connect this literature to the zero-torque counterfactual (ZTCF) and zero-velocity counterfactual (ZVCF) decompositions implemented in the UpstreamDrift platform, which separate, at every instant of a measured swing, the acceleration attributable to accumulated motion (the drift term) from the acceleration attributable to the applied joint torques (the control term). We formalize the “retain-early / release-late” hypothesis in counterfactual terms, inventory the repository’s models and energy-transfer computations, identify the measurement gaps (notably segmental energy-flow accounting), and specify falsifiable simulation experiments — torque-timing sweeps, counterfactual energy budgets, re-simulation parity — that can decide the question quantitatively. No new experimental data are reported; every quantitative claim is sourced to the cited literature or to inspectable code.

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i Provenance and scope

Deliverable of UpstreamDrift epic [#8373](#) (sub-issues [#8374](#)–[#8379](#)). A *review and modeling-perspective* paper: no new experimental data are reported; the experiments of Section 9 are planned work tracked in issue [#8377](#). All references were verified against publication records at the time of writing; repository links point to the **main** branch of [D-sorganization/UpstreamDrift](#).

1 Introduction

Clubhead speed at impact is the dominant controllable determinant of drive distance, and the golf downswing is one of the most-studied examples of a *sequenced* multi-segment rotation in sport biomechanics ([Hume, Keogh, and Reid 2005](#); [McPhee 2022](#)). Across measurement systems and populations, the downswing of a skilled golfer shows a characteristic order: the pelvis reaches its peak angular speed first, then the thorax, then the lead arm, and finally the club, with each successive segment reaching a higher peak speed than the one before it ([Cheetham et al. 2008](#); [Tinmark et al. 2010](#)). This pattern — the *kinematic sequence*, an instance of the general proximal-to-distal (P→D) sequencing observed in throwing and striking skills ([Putnam 1993](#)) — is routinely interpreted as the visible signature of mechanical energy being generated in the large proximal segments and transferred outward to the small, fast distal ones.

That interpretation immediately raises a question that the descriptive literature does not settle:

Should a golfer attempt to transfer energy from proximal to distal segments *early* in the downswing, or should energy be *retained* in the proximal segments early and

the transfer deliberately *accelerated* late?

The question is not academic. The two strategies imply opposite coaching cues (“fire the club from the top” versus “hold the lag and let it release”), opposite torque programs, and different predictions for what counterfactual decompositions of a measured swing should show. It also connects to a long-standing puzzle: double-pendulum analyses have argued since the 1960s that the highest clubhead speeds arise when the distal joint is *not* driven early — when release is passively or actively delayed (Cochran and Stobbs 1968; T. Jorgensen 1970; Milburn 1982; Pickering and Vickers 1999) — while the coaching translation of the kinematic sequence often emphasizes initiating distal motion promptly after transition.

The UpstreamDrift project approaches this question with *counterfactual dynamics*. Two decompositions are implemented across its physics engines:

- the **zero-torque counterfactual (ZTCF)** — the acceleration the system would exhibit at the current instant if all applied joint torques were removed, isolating what accumulated motion, gravity, and inter-segment coupling produce on their own; and
- the **zero-velocity counterfactual (ZVCF)** — the acceleration the applied torques, together with gravity, would produce at the current pose if the system were momentarily at rest, removing all motion-dependent effects (the pure torque contribution is the control term of the drift-control split defined in Section 6).

Preliminary ZTCF/ZVCF analyses with the repository’s MATLAB pipeline (Section 6) motivated the hypothesis examined here: that *restraining* the P→D handoff early in the downswing — keeping kinetic energy in the proximal segments and in the arms-plus-club system rotating as a quasi-rigid unit — and accelerating the handoff late may be the more effective strategy. This article evaluates that hypothesis against the published literature and specifies how to test it quantitatively with the tools already in the repository.

Contributions.

1. a synthesis of the mechanics of sequenced swings, from the summation-of-speed principle to segment-interaction dynamics and parametric energy transfer (Section 2);
2. a methodological account of what “energy transfer” means operationally — joint power, joint-force power, segmental energy accounting, and induced accelerations (Section 3);
3. a critical review of the empirical evidence on where P→D transfer occurs and how it differs between skilled and less-skilled players (Section 4);
4. a formalization of the early-versus-late transfer question, including the ZTCF/ZVCF counterfactual framework and the specific predictions the “retain-early / release-late” hypothesis makes (Section 5, Section 6); and
5. an inventory of the models and computations available in UpstreamDrift today, an honest gap analysis, and a set of falsifiable simulation experiments that would decide the question (Section 7, Section 9).

2 Mechanics of sequenced swings

2.1 The summation-of-speed principle and its limits

The oldest formalization of P→D sequencing is the *summation of speed* principle: in skills demanding high end-point speed, each segment should begin its motion at the moment of peak speed of the segment proximal to it, so that distal speed is “built on top of” proximal speed (Bunn 1972). Milburn applied the principle directly to golf with a double-pendulum analysis of the downswing and found the characteristic pattern of an initially dominant proximal (arm) rotation followed by rapid distal (club) rotation, noting that “*an initial delay in the uncocking of the wrist*” allows the arm to reach greater acceleration and the club’s acceleration to be summed with the proximal segment’s maximum (Milburn 1982). Note the structure of that finding: the summation principle, taken naively, prescribes *sequential initiation*; Milburn’s analysis showed the benefit comes specifically from *delaying* the distal contribution.

Putnam’s segment-interaction analyses put the principle on a dynamical footing and exposed its limits (Putnam 1991, 1993). In a linked chain, each segment’s angular acceleration depends not only on joint torques

but on *motion-dependent interaction torques* — centripetal and Coriolis terms proportional to squares and products of segment angular velocities. Putnam showed the P→D pattern in striking and throwing arises substantially from these interaction torques: proximal deceleration and distal acceleration late in the motion are coupled through the joint force and occur even without a distal joint torque. Two implications matter for golf. First, kinematic order does not reveal *which* torques produced it: an identical peak sequence can arise from different mixes of muscular and interaction torques. Second, the proximal deceleration observed in experts need not be an active “braking” strategy — it can be the dynamical *consequence* of the distal segment accelerating and back-reacting on the chain. Marshall and Elliott further cautioned that planar peak-ordering analyses miss long-axis rotations, which peak very late in racquet skills and contribute substantially to end-point speed (Marshall and Elliott 2000); the golf analogue is forearm rotation and wrist action in the final phase of the downswing.

2.2 Double-pendulum mechanics of the golf downswing

The two-lever, one-hinge model of the downswing — an “arm” link rotating about a fixed or moving hub, connected by a wrist hinge to a “club” link — has been the workhorse of golf mechanics since Cochran and Stobbs’ *Search for the Perfect Swing* (Cochran and Stobbs 1968) and Jorgensen’s dynamical analysis fitted to stroboscopic photographs of a professional swing (T. Jorgensen 1970; T. P. Jorgensen 1999). Its enduring value here is that it makes the *timing* of the P→D handoff an explicit variable (the “release”: the point at which the wrist angle is allowed, or driven, to open) and exposes the mechanism by which delay pays. Three results anchor the analysis:

Delayed release increases clubhead speed. Pickering and Vickers, sweeping release angles in a driven double-pendulum model, found clubhead velocity at impact increases as release is delayed — the classical “late hit” (Pickering and Vickers 1999). Jorgensen’s fitted model similarly reproduced the expert swing only with a substantially delayed opening of the wrist angle (T. Jorgensen 1970; T. P. Jorgensen 1999). In these models an *early* release (early P→D handoff) is strictly inferior: club kinetic energy purchased early is paid for with arm speed that never develops.

Late wrist torque helps; early wrist torque is the wrong tool. Sprigings and Neal, using a torque-driven three-segment model with physiologically bounded actuators, found that an active wrist torque applied *during the latter stages of the downswing* increased clubhead speed by approximately 4% (Sprigings and Neal 2000); Sprigings and MacKenzie confirmed the delayed-release benefit in a follow-up simulation focused specifically on the timing of release (Sprigings and MacKenzie 2002). The point is not that distal effort is useless, but that its productive window is late.

Energy transfer to the club can be largely passive — and is governed by the changing geometry. White analyzed an *undriven* double pendulum and showed that the transfer of kinetic energy from arms to club late in the downswing is a parametric-transfer phenomenon: as the club swings out, the effective moment of inertia of the system changes, and conservation of energy and angular momentum move kinetic energy distally without any wrist torque at all (White 2006). The efficiency of this transfer is controlled principally by the wrist-cock angle maintained before release (White 2006). Miura’s analysis of the “inward pull” complements this: an expert’s hands moving toward the body near impact — equivalent to shrinking the hub radius when centrifugal loading is fully developed — performs work against the centrifugal force that appears as additional clubhead kinetic energy (Miura 2001). Nesbit and McGinnis’ hub-path analyses generalize the same idea: the non-circular hand path of skilled golfers, including its late-downswing tightening, is a mechanism for shaping kinetic transfer from golfer to club (Nesbit and McGinnis 2009; Nesbit and McGinnis 2014).

Together these results describe a machine whose highest-output mode is: *load the proximal rotation first, hold the distal joint folded (retaining energy in a compact, fast-rotating configuration), then let geometry and — in the last phase — modest distal torque move the energy outward rapidly.* The double-pendulum literature is already an argument for **retain-early / release-late**.

2.3 Optimal control and torque sequencing

If the timing question is posed as an optimization — what torque program maximizes clubhead speed (or carry) subject to actuator limits? — the answer has been studied repeatedly. Lampsas’s early optimal-control study of a driven double pendulum computed torque histories maximizing drive distance, and notably found the *optimal* torques differed in structure from those measured (via inverse dynamics) in his own swing (Lampsas 1975) — early evidence that human swings need not be optimal and that the optimum is a meaningful target rather than a redescription of expert behavior. Sharp, fitting shoulder-arm-club models to expert swing data and then optimizing torque programs, established a pattern for the best use of the available driving torques across the downswing phases, with the distal contribution structured late rather than distributed evenly (Sharp 2009). Three-dimensional forward-dynamics models extended the approach with anatomically meaningful actuators and flexible shafts: MacKenzie and Sprigings’ 3D model (MacKenzie and Sprigings 2009), the carry-optimized model of Balzerson, Banerjee and McPhee (Balzerson, Banerjee, and McPhee 2016), and the six-degree-of-freedom dynamic optimization of McNally and McPhee (McNally and McPhee 2018) all generate expert-like clubhead speeds with torque programs whose distal elements act late in the downswing. McPhee’s review surveys this modeling lineage (McPhee 2022).

A caution cuts the other way: optimal control optimizes *within a model*, and the sensitivity of optimal timing to model structure (hub mobility, shaft flexibility, actuator rate limits) is exactly why a multi-engine platform with cross-engine validation (Section 7) is the right instrument for revisiting the question.

3 What “energy transfer” means operationally

Claims about energy transfer require an accounting framework. Four levels of analysis appear in the literature, and they are not interchangeable.

3.1 Joint power and work

For a joint spanning segments i and $i+1$ with net joint torque τ and relative angular velocity ω , the *joint (moment) power* is $P_m = \tau \omega$; its time integral over a phase is the mechanical work done by the net moment. Positive work indicates generation (or concentric action), negative work absorption. This is the quantity most golf studies report, and it is the basis of the work-power budget of Nesbit and Serrano (Section 4.1) (Nesbit and Serrano 2005).

3.2 Joint-force power and segmental energy accounting

Joint moments are not the only conduit of mechanical energy: the joint reaction force $\mathbf{F}$ acting at a joint moving with velocity $\mathbf{v}$ transmits *joint-force power* $P_f = \mathbf{F} \cdot \mathbf{v}$ from one segment to the next. Robertson and Winter’s classic formulation balances, for each segment, the rate of change of segment mechanical energy against the sum of joint-force powers and moment powers at its proximal and distal ends, distinguishing energy *transferred between* segments from energy *generated or absorbed* at joints (Robertson and Winter 1980). This is the correct level of analysis for the question posed here, because the “retain-early / release-late” hypothesis is precisely a claim about the *time course of inter-segment transfer terms*, not about peak angular velocities. As Section 7.6 notes, this accounting is the principal measurement gap in the UpstreamDrift Python stack today (the MATLAB pipeline’s per-joint linear-plus-angular work decomposition is its closest existing realization).

3.3 Induced accelerations and dynamic coupling

In a coupled chain, any single torque induces accelerations at *every* joint — including joints it does not span — through the inverse inertia matrix (Zajac and Gordon 1989). Induced-acceleration and power-transfer analyses in locomotion research (Zajac, Neptune, and Kautz 2002) show that intuitions of the form “muscle X accelerates joint Y” routinely fail in coupled systems. For golf this matters because a torso torque early in the downswing partly accelerates the *club* (through coupling), and a wrist torque late in the downswing back-reacts on the torso; the net energy routing is an emergent property of the whole chain.

The ZTCF/ZVCF decompositions of Section 6 are the counterfactual cousins of this analysis: rather than attributing acceleration to individual torques, they split the total acceleration into a motion-dependent (drift) part and a control part, instant by instant.

3.4 Kinematic sequence as a proxy — and its failure modes

Peak-ordering of segment angular speeds is a *proxy* for energy flow, not a measurement of it. Marsan and colleagues showed that the identified kinematic sequence in golf depends materially on which angular-velocity component (or norm, or projection) is analyzed: thirteen golfers analyzed with seven component-selection methods yielded different sequences (Marsan et al. 2019). Marshall and Elliott’s long-axis-rotation caution (Marshall and Elliott 2000) is a special case of the same problem. Any serious test of the timing hypothesis must therefore be run at the level of energy and power accounting (Section 3.2), with the kinematic sequence reported as corroborating description only.

4 Empirical evidence in golfers

4.1 The work and power budget of the swing

Nesbit’s full-body kinematic-kinetic study and the companion work-power analysis of Nesbit and Serrano remain the most complete published energy budgets of the swing (Nesbit 2005; Nesbit and Serrano 2005). Using subject-specific full-body models of four golfers of widely differing skill, they report that the trunk complex (lumbar and thoracic spine plus hips) contributes 68.7–72.2% of total swing work, the shoulder and arm joints 24.3–28%, and the legs only 3.3–3.8% (Nesbit and Serrano 2005). Two further observations from the same study bear directly on this article’s question. First, power generation and clubhead speed decreased systematically with increasing handicap. Second, the authors’ phase analysis locates the *generation* of energy predominantly in the earlier downswing (torso-dominated) and the *conversion* into clubhead speed late (shoulder/arm and club interface) — i.e., generation and transfer are temporally separated, which is a precondition for the retain-early / release-late strategy to even be possible.

The ground is the ultimate source of the external impulse. Han and colleagues characterized golfer-ground interaction in 63 skilled golfers and found meaningful associations between ground-interaction force and moment parameters and maximum clubhead speed (Han et al. 2019). Recent work has pushed this to an explicitly energetic formulation: in thirty golfers (fifteen professionals, fifteen high-level amateurs), foot-ground interaction variables alone explained 35.5% of clubhead-speed variance, rising to 75.4% when trunk sequencing and an *impulse-based energy-transfer efficiency* measure (effectiveness of mechanical impulse transmission from proximal segments to the club) were added; mediation analysis attributed the dominant indirect pathway to transfer efficiency rather than to sequencing per se (“Foot-Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing” 2026). That last distinction — transfer *efficiency* mattering where sequence *order* alone does not — is a recurring theme of the skill-level literature reviewed next.

4.2 Kinematic-sequence evidence and skill-level differences

The descriptive P→D findings are robust in skilled populations. Tinmark and colleagues, studying elite golfers across full and partial shots, found a significant proximal-to-distal temporal ordering of peak speeds (pelvis → torso → hands/club) with successive amplification of peak speed at every level of effort, in both men and women (Tinmark et al. 2010). Cheetham and colleagues compared amateur recreational players with touring professionals on transition- and downswing-phase kinematic sequence parameters and found professionals showed higher peak-speed magnitudes and more consistent sequencing/timing characteristics (Cheetham et al. 2008). Zheng and colleagues’ three-dimensional comparison of professional and amateur swings found systematic temporal and spatial differences, with amateurs deviating progressively more from professional norms as handicap increased (Zheng et al. 2008). Meister and colleagues provide elite rotational benchmarks (pelvis-thorax separation and its timing) against which amateurs can be compared (Meister et al. 2011), and Lindsay, Mantrop and Vandervoort review the broader catalogue of skill-level biomechanical differences (Lindsay, Mantrop, and Vandervoort 2008).

Three refinements prevent an overly tidy reading:

1. **Sequence order is common but not universal — even among the best.** Skilled golfers whose peak ordering deviates from the canonical pelvis-thorax-arm-club pattern are documented, and recent work examines how alternative sequence patterns among elite players relate to clubhead speed (Lee, Ehrlich, and Cain 2026). On-plane analyses of the “axle-chain” system in 66 skilled golfers found clubhead-speed associations concentrated in specific angular-motion characteristics rather than in gross ordering (Madrid et al. 2020).
2. **Within-player timing differences do not straightforwardly explain shot quality.** Comparing well-timed and mistimed shots *within* highly skilled players, Neal and colleagues found no significant differences in the timing lags between peak angular speeds of contiguous segments (Neal et al. 2007) — the felt sense of “timing” lives somewhere other than the inter-peak lags of a five-segment model.
3. **Method sensitivity.** As noted in Section 3, the identified sequence depends on component-selection methodology (Marsan et al. 2019), so cross-study comparisons of ordering statistics carry real uncertainty.

What survives these refinements: *skilled golfers generate larger proximal speeds, separate transition from release more distinctly, and transmit impulse/energy to the club more efficiently; merely exhibiting a $P \rightarrow D$ peak order is neither unique to skill nor sufficient for it* (Cheetham et al. 2008; Zheng et al. 2008; “Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing” 2026; Lee, Ehrlich, and Cain 2026).

4.3 Where the transfer occurs

Synthesizing Section 4.1 and Section 4.2 with Section 2: in a skilled swing, the dominant energy *generation* occurs in the trunk-hip complex during the first two-thirds of the downswing (Nesbit and Serrano 2005); the dominant *transfer to the club* occurs in the final phase, through the wrist interface, by a mechanism that is substantially parametric/geometric (wrist-cock release under centrifugal loading, hub-path tightening) with a modest late active wrist contribution (White 2006; Miura 2001; Sprigings and Neal 2000; Nesbit and McGinnis 2009). Less-skilled players truncate exactly this separation: early release (“casting”) moves energy into the club before proximal speed has peaked, producing the lower distal peak speeds and transfer ratios observed in amateur cohorts (Cheetham et al. 2008; Zheng et al. 2008; Milburn 1982).

5 The timing question, stated precisely

5.1 Two strategies

Let $E_p(t)$ be the kinetic energy of the proximal subsystem (trunk and arms, with the club riding folded) over a downswing $t \in [t_0, t_{\text{imp}}]$, and define the transfer functional $P_w(t)$ as the inter-segment power crossing the wrist interface (joint-force power plus moment power, in the sense of Section 3.2). The two strategies are:

- **S1 (transfer early):** make $P_w(t)$ large as soon after transition as possible; the club begins acquiring its final energy early.
- **S2 (retain early, release late):** keep $P_w(t)$ small early (the wrist transmits centripetal force but little power) while proximal torques maximize E_p ; then allow/drive a large P_w concentrated in the final phase.

The double-pendulum, delayed-release and optimal-control literatures reviewed in Section 2 uniformly favor S2 within their model classes (Milburn 1982; Pickering and Vickers 1999; Sprigings and Neal 2000; Sprigings and MacKenzie 2002; Sharp 2009; White 2006). The mechanistic reasons compose into a coherent account:

1. **Inertia management.** With the wrist cocked, the system’s moment of inertia about the hub is small, so proximal torque buys angular velocity cheaply; early release raises the inertia while proximal torque is still accelerating the system (White 2006; T. P. Jorgensen 1999).

2. **Actuator physiology.** Torque-velocity limits mean the wrist’s small muscles add meaningful power only while wrist angular velocity is modest; early wrist torque squanders that window and can even require *negative* (restraining) torque later (Sprigings and Neal 2000).
3. **Parametric transfer needs developed centrifugal loading.** The passive outward energy flow at release, and the inward-pull gain, scale with the centrifugal field established by proximal rotation — weak early, strong late (White 2006; Miura 2001).
4. **Interaction torques do the late work for free.** Motion-dependent torques accelerate the distal segment and decelerate the proximal one automatically once release begins; the golfer’s job is to *time* this cascade, not to muscle it (Putnam 1993).

Empirically, S2-like signatures separate skill groups (Section 4.2), while the within-expert null result on inter-peak lags (Neal et al. 2007) suggests that once S2 is grossly achieved, residual performance variance lives in finer variables (release profile shape, hub-path geometry, impulse-transfer efficiency) — exactly what the counterfactual framework below is designed to expose.

5.2 What would falsify S2?

To keep the question scientific rather than rhetorical, we state what evidence would count against the retain-early/release-late hypothesis:

- an optimal-control solution, in a validated 3D model with physiological actuator limits, whose optimal wrist-interface power is front-loaded;
- measured expert swings in which early wrist-interface power (normalized) is *positively* associated with clubhead speed across players;
- counterfactual (ZTCF/ZVCF) analyses showing that in faster swings the control term dominates club acceleration *early* rather than late.

The proposed experiments of Section 9 operationalize these tests.

6 Counterfactual decompositions: ZTCF and ZVCF

6.1 Definitions

Write the chain’s equation of motion in standard manipulator form:

$$M(q)\ddot{q} = \tau - b(q, \dot{q}), \quad b(q, \dot{q}) = C(q, \dot{q})\dot{q} + g(q) + d(\dot{q}), \quad (1)$$

with inertia matrix M , applied generalized torques τ , Coriolis and centripetal terms $C\dot{q}$, gravity g , and velocity-dependent dissipation d . Two counterfactual accelerations are defined *pointwise* at each measured state $(q, \dot{q}, \tau)$ of a swing:

$$\ddot{q}_{\text{ZTCF}} = M(q)^{-1}(-b(q, \dot{q})), \quad (2)$$

$$\ddot{q}_{\text{ZVCF}} = M(q)^{-1}(\tau - b(q, \mathbf{0})) = M(q)^{-1}(\tau - g(q)), \quad (3)$$

where the simplification in Equation 3 holds because the Coriolis term (quadratic in velocity) and viscous dissipation (linear in velocity) vanish at $\dot{q} = \mathbf{0}$. The ZTCF (Equation 2) answers “*what would the system do right now if every actuator switched off?*” — it is the **drift** field of the affine control system, containing gravity plus all motion-dependent interaction dynamics. The ZVCF (Equation 3) answers “*what would the currently applied torques, together with gravity, produce from a standstill in this pose?*” — it removes every motion-dependent term but is *not* the pure control contribution, since gravity remains in Equation 3; the pure torque term is $M^{-1}\tau$ in the split below. Because Equation 1 is affine in τ , the actual acceleration superposes exactly:

$$\ddot{q} = \underbrace{M^{-1}(-b(q, \dot{q}))}_{\text{drift} = \text{ZTCF}} + \underbrace{M^{-1}\tau}_{\text{control}}, \quad (4)$$

an identity enforced to numerical tolerance by the repository’s test suite (Section 7.5).

6.2 Two operational variants

The repository implements two distinct operationalizations, and the difference matters for interpretation:

Pointwise (instantaneous) counterfactuals. The engine-agnostic Python implementation ([simulation_backends/ztcf_](#)) evaluates Equation 2 and Equation 3 at each sample of an already-measured trajectory — explicitly *not* a forward-integrated rollout. Each sample answers a local “what if”; this is the right tool for drift/control *time profiles* along a real swing.

Re-simulation counterfactuals. The original MATLAB/Simscape pipeline ([run_ztcf_simulation.m](#)) re-simulates the full model with a torque “killswitch” applied at time t , for a sweep of t values, tabulating the counterfactual state reached with torques off. The pipeline forms the difference table $\text{DELTA} = \text{BASE} - \text{ZTCF}$ ([process_data_tables.m](#)), attributing to active control everything the passive system would not have done. A companion ZVCF model applies the measured torques to the same pose with velocities zeroed ([SCRIPT_ZVCF_GENERATOR.m](#)); the ZVCF model is run without gravity so that, when its samples are set against the ZTCF (which retains gravity), the gravitational contribution is not double-counted. Note that this decomposition is meaningful *pointwise*, at matched states, in the sense of Equation 4: the ZVCF script records one sample per zero-velocity pose, and separately integrated counterfactual rollouts do not superpose at the trajectory level in a nonlinear multibody system, because M and b evolve differently once the counterfactual state diverges. The MATLAB pipeline additionally computes, for every joint and every table (BASE, ZTCF, DELTA), the linear work $\int \mathbf{F} \cdot \mathbf{v} dt$, angular work $\int \tau \cdot \omega dt$, total work and power, and the fractional contributions ZTCF/BASE and DELTA/BASE ([calculate_work_impulse.m](#), [calculate_total_work_power.m](#)) — precisely the Robertson-Winter-style joint-interface accounting (Section 3.2) that the timing question needs, including a kinematic-sequence plot computed *separately for the BASE, ZTCF and DELTA components* ([SCRIPT_324_PLOT_ZTCF_KinematicSequence.m](#)).

6.3 The counterfactual form of the timing hypothesis

The ZTCF/ZVCF language makes the S2 hypothesis crisp. Along a measured downswing, project the drift and control accelerations onto the club’s speed-producing direction and integrate the associated powers. S2 predicts, for high-performing swings:

- **Early downswing:** the *control* term should dominate energy input at proximal joints, while at the wrist the control contribution to club angular acceleration should be small or negative (restraining lag against the developing centrifugal field). The club’s ZTCF profile should show the drift field already “wants” to open the wrist — control is *holding energy back*.
- **Late downswing:** the *drift* term (parametric transfer, centrifugal release) should dominate club acceleration, with a brief late window of positive wrist control power ([Springs and Neal 2000](#)). The DELTA (active) share of club-side work should concentrate before release; the ZTCF (passive) share should account for most club energy gain after release ([White 2006](#)).

This pattern is stated here as a *hypothesis to be quantified*, not a result: the substantiating figures and tables are the deliverable of issue [#8377](#). Conversely, if high-performing swings show early-dominant wrist control power, S2 is falsified in exactly the terms of Section 5.

7 What UpstreamDrift computes today

This section inventories the repository’s relevant capabilities; links point to the `main` branch.

7.1 Models

Table 1: Golf swing models available in the repository.

Model	Location	Notes
Double pendulum (arms + club, clubhead tip mass)	pendulum_simulator/physics.py	Lagrangian dynamics; viscous + Coulomb friction; analytic M , C , g
Triple pendulum	pendulum_simulator/physics_trip.py	Three-segment planar chain
8-DOF closed-loop golfer (two arms + shared club)	pendulum_simulator/physics_golfer.py	Holonomic loop constraints, constraint-consistent dynamics; JAX variant
Backend-neutral golf model (ODE / MuJoCo / MJX / Warp)	simulation_backends/	One parameter set rendered to every backend; shared trace schema
MuJoCo humanoid golfer	mujoco_humanoid_golf/	Full-body model with club
Simscape 2D/3D golf models	Simscape_Multibody_Models/	Source of the original ZTCF/ZVCF methodology
Flexible shaft (FEM)	physics/flexible_shaft.py	Beam model; strain energy not yet in the energy budget

7.2 Counterfactual stack

The normative contract lives in the engine-core dynamics interface ([_dynamics_interface.py](#)): `compute_drift_acceleration`, `compute_control_acceleration`, `compute_ztcf`, `compute_zvcf`, with the superposition identity (Equation 4) stated as a contract. Engine implementations exist for MuJoCo, Drake, Pinocchio, OpenSim, MyoSuite, the analytic pendulum engines, and the Simscape bridge; the capability matrix is documented in [engine_capabilities.md](#). On top of these sit:

- the pointwise engine-agnostic evaluator with HDF5 persistence ([ztcf_zvcf.py](#));
- a MuJoCo delta-based counterfactual analyzer reporting torque-attributed and velocity-attributed acceleration effects ([counterfactuals.py](#));
- a Newton-Euler ZTCF force module with an analytic single-pendulum validation identity ([biomechanics/ztcf.py](#));
- induced-acceleration analyzers (gravity / drift / control / total components) ([induced_acceleration.py](#));
- an orchestration layer with capability gating and a REST/UI surface ([analysis/orchestrator.py](#));
- the MATLAB re-simulation pipeline described in Section 6.

7.3 Energy and power computations

- **Inter-segment power flow (MuJoCo):** [power_flow.py](#) computes per-DOF joint powers $\tau\omega$, work decomposed into drift and control shares, per-body kinetic and potential energies via center-of-mass Jacobians, per-body parent/child transfer bookkeeping, and an energy-conservation residual derived from the equations of motion.
- **Joint work/power/impulse metrics (engine-agnostic):** [power_work_metrics.py](#) (positive/negative/net work, generation-absorption splits, impulse, quasi-stiffness); [energy_metrics.py](#) (system energy summaries).
- **Conservation monitoring:** [energy_monitor.py](#) enforces drift bounds on total energy during integration (note: its potential-energy term is a documented first-order approximation, adequate for drift detection only).
- **MATLAB energetics:** the per-joint linear/angular work and power decomposition across BASE/ZTCF/DELTA tables (Section 6) — currently the richest joint-interface energy accounting in the repository.

7.4 Sequencing analytics

- [kinematic_sequence.py](#): peak detection, ordering consistency against a caller-supplied expected order, timing gaps, distal/proximal speed-gain ratios, post-peak deceleration rates. (Deliberately method-neutral; the expected order is user-defined.)
- [coordination_metrics.py](#): vector-coding coupling angles, continuous relative phase, pairwise lag matrices — quantitative sequencing measures less brittle than peak-ordering.
- Plot renderers for sequence traces, peak-timing Gantt charts, and lag matrices ([_coordination_sequence.py](#)).
- An approved internal research review of the kinematic-sequence metric, including literature benchmarks and uncertainty-reporting guidance ([kinematic_sequence.md](#)).

7.5 Validation evidence

The counterfactual stack is contract-tested: the pointwise evaluator reproduces the analytic double-pendulum ground truth to 10^{-9} via the ODE and analytical providers, and agrees with the independent MuJoCo backend to 10^{-7} when that engine is available ([test_ztcf_zvcf.py](#)); drift-plus-control reconstructs the measured acceleration ([test_counterfactual_experiments.py](#)); work-energy-theorem and conservation-law property tests run in CI ([tests/integration/conservation_laws/](#)); and a cross-engine comparison service produces divergence-annotated ZTCF/ZVCF panels between backends ([cross_engine_comparison.md](#)). The project’s design guidelines make the drift-control decomposition and ZTCF/ZVCF availability normative across engines ([project_design_guidelines.qmd](#)).

7.6 Gaps relevant to this article

Table 2: Known gaps (tracked in issue [#8378](#)).

Gap	Consequence for the timing question
Two ZTCF operationalizations (pointwise vs re-simulation) coexist; no Python re-simulation ZTCF	Analyses must state which variant they use; parity between variants is untested
Inter-segment power transfer implemented only for MuJoCo, as joint power $\tau\omega$	Wrist-interface <i>force</i> power $\mathbf{F} \cdot \mathbf{v}$ — central to S1/S2 — is not yet computed in Python
No Robertson-Winter segmental energy-flow accounting in Python	The transfer functional $P_w(t)$ must currently come from the MATLAB tables
<code>ConservationMonitor</code> potential energy is first-order approximate	Do not report absolute PE from it; use engine energy accessors
Flexible-shaft strain energy absent from the energy budget	Late-downswing shaft recoil energy is invisible to current accounting
Proximal-braking-efficiency metric is an open TODO	Deceleration-side claims must rely on generic post-peak deceleration rates

8 Synthesis: early or late?

What is established. (i) Skilled swings exhibit P→D ordering with successive speed amplification across shot types and populations ([Tinmark et al. 2010](#); [Cheetham et al. 2008](#)). (ii) The trunk-hip complex does most of the work; the club acquires most of its energy late, through the wrist interface ([Nesbit and Serrano 2005](#); [Nesbit 2005](#)). (iii) In every published double-pendulum, delayed-release, and optimal-control analysis, delaying the distal handoff increases clubhead speed, and productive distal torque is late ([Milburn 1982](#); [Pickering and Vickers 1999](#); [Springs and Neal 2000](#); [Springs and MacKenzie 2002](#); [Sharp 2009](#)). (iv) The late transfer is substantially passive — parametric/geometric — once proximal rotation has established the centrifugal field ([White 2006](#); [Miura 2001](#)). (v) Skill differences appear in magnitudes, transition timing, and transfer efficiency more than in gross order ([Cheetham et al. 2008](#); [Zheng et al. 2008](#); “Foot–Ground

Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing” 2026).

What this implies for the question. The weight of evidence supports **S2: retain energy proximally early in the downswing and accelerate the transfer late**. The phrase “retain energy proximally” deserves precision: it does not mean suppressing proximal motion — proximal segments should be accelerated as hard as the movement budget allows — it means *deferring the handoff*: keeping the wrist interface transmitting force but little power (club folded, riding with the arms) until late, so that the energy pool is large and the centrifugal field strong when the parametric transfer is allowed to run. In counterfactual terms: early control effort should appear at proximal joints and, at the wrist, partly as *restraint* of the drift field’s tendency to release; late club energy gain should be drift-dominated with a brief late window of positive wrist control power (Section 6).

What remains genuinely open. (a) The published S2 evidence is model-based or observational; a *direct* energetic accounting of the wrist interface across skill levels (the $P_w(t)$ time course) has not been reported in the literature reviewed here — the closest is the impulse-transfer-efficiency result of (“Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing” 2026), which is consistent with, but does not resolve, the timing profile. (b) Optimal timing is plausibly equipment- and anthropometry-dependent, and model sensitivity is untested across engines. (c) Whether *actively* restraining release (negative wrist torque early) outperforms merely *not driving* it is unresolved in 3D models with realistic actuators — 2D analyses suggest passive holding suffices (T. P. Jorgensen 1999; White 2006), while actuator-based models leave room for an active-restraint benefit (Sprigings and Neal 2000). These are the questions the repository’s toolchain can answer.

9 Proposed experiments

Each design below has a pre-registered, falsifiable outcome measure. All are tracked in issue #8377; no results exist yet, and none are reported here.

E1 — Torque-timing sweep (double pendulum and 8-DOF golfer). Drive the models of Table 1 with parameterized torque programs: proximal torque fixed; wrist-torque onset time t_{on} , sign profile (restrain-then-drive vs drive-only), and magnitude swept on a grid. Outcome: clubhead speed at impact versus t_{on} and profile. S2 predicts a late-interior optimum of t_{on} and a non-inferior restrain-then-drive profile. The MJX/Warp batched backends make the sweep cheap; the ODE backend is the reference.

E2 — Counterfactual energy budget along swept trajectories. For each E1 trajectory, evaluate the pointwise ZTCF/ZVCF and drift/control split (Equation 4) and integrate drift-side and control-side club power. Outcome: the drift-vs-control share of final club kinetic energy in early vs late downswing halves, as a function of performance. S2 predicts the drift share of late club energy gain rises with clubhead speed.

E3 — Re-simulation parity. Implement the killswitch re-simulation ZTCF in Python on the backend-neutral model; compare against the pointwise ZTCF profiles and the MATLAB pipeline on matched trajectories. Outcome: documented agreement/divergence between the two ZTCF operationalizations, closing the top gap of Table 2.

E4 — Wrist-interface power accounting. Implement Robertson-Winter segmental energy flow (joint force power plus moment power per interface; issue #8378), then compute $P_w(t)$ for E1’s best and worst performers. Outcome: the direct time course of the transfer functional defined in Section 5 — the primary evidence either strategy needs.

E5 — Mocap replication. Run the E2/E4 analyses on measured swings ingested through the motion pipeline (C3D → tracked motion) across skill levels. Outcome: observational test of the counterfactual predictions of Section 6 against human data, using the uncertainty-reporting conventions of the internal research review.

10 Limitations

This article synthesizes literature and specifies methodology; it reports no new data, and the S2 hypothesis remains unquantified in the repository’s own terms until the experiments of Section 9 are executed. The kinematic-sequence evidence base carries methodological sensitivity (Marsan et al. 2019); several classical results rest on planar models whose generalization to 3D swings with mobile hubs is a live question (McPhee 2022); and the work-budget percentages cited from (Nesbit and Serrano 2005) derive from four subjects. One recent source (“Foot–Ground Interaction and Clubhead Speed: Impulse-Based Energy Transfer as the Key Mechanism in the Golf Swing” 2026) is cited without its author list because the publisher’s site was unreachable through this environment’s network proxy; resolve the DOI for full details. Finally, the arguments here concern clubhead *speed*; accuracy, consistency, and injury-risk trade-offs of late-transfer strategies (Hume, Keogh, and Reid 2005; Lindsay, Mantrop, and Vandervoort 2008) are outside scope.

11 Conclusion

Proximal-to-distal sequencing in the golf swing is real, robust, and — as a coaching target — chronically over-interpreted. The mechanical literature, from Cochran and Stobbs through modern optimal control, does not support “pass the energy along as soon as possible.” It supports a subtler program: *build the energy proximally with the club held folded, defer the handoff, and let the transfer run — mostly parametrically, with a brief late assist — when the centrifugal field can pay for it.* Skilled players differ from amateurs along exactly this axis: not in whether their segments peak in order, but in how much energy the proximal system holds at transition and how late and how efficiently it is handed to the club. The ZTCF/ZVCF framework turns this qualitative account into measurable time profiles — drift versus control, early versus late — and the experiment suite specified here (epic #8373) is designed to quantify the retain-early/release-late advantage or falsify it. Either outcome would be a contribution.

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